

Samir Shrestha

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PROFESSIONAL SUMMARY

AI & Machine Learning graduate student with 5 years of professional software development experience building production-ready backend and full-stack applications. Strong foundation in Python, JavaScript, PHP, and SQL, with experience developing REST APIs, database-driven systems, and scalable web applications. Hands-on experience with machine learning workflows, dataset preparation, model evaluation, and reproducible experimentation using PyTorch and TensorFlow. Proven ability to collaborate across teams, improve system reliability, and deliver maintainable solutions in real-world environments. Seeking full-time opportunities in software engineering, backend development, AI/ML engineering, and related technology roles.

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript, PHP, SQL
- Machine Learning & AI: Supervised learning, unsupervised learning, neural networks, CNNs, RNNs, transformers, hyperparameter tuning, model optimization, cross-validation, model evaluation
- Deep Learning Frameworks: PyTorch, TensorFlow, Scikit-learn
- Natural Language Processing: Text preprocessing, tokenization, embeddings, sequence modeling, transformer architectures, autoregressive models
- Web Development: React, Next.js, Vue.js, Node.js, Laravel
- Backend & Systems: REST APIs, relational databases, SQL optimization, Linux, Docker, AWS, distributed systems fundamentals
- Data & ML Engineering: Data cleaning, feature engineering, dataset preparation, ML pipelines, experiment tracking, reproducible workflows, validation
- Tools & Practices: Git, debugging, code reviews, production deployments, documentation

EXPERIENCE

SOFTWARE DEVELOPER — MANDALA TECH (AUGUST 2022 – APRIL 2025)

Built and maintained backend services and data-driven web applications using PHP (Laravel), JavaScript, and SQL in a production environment.

- Developed and maintained REST APIs supporting frontend-backend integration and business workflows
- Designed and optimized relational database schemas and SQL queries to improve performance and data access efficiency
- Built reusable backend modules and services to support scalable application architecture
- Implemented reporting features and system integrations for business-critical applications
- Improved system reliability through debugging, refactoring, and performance optimization of production systems
- Contributed to production deployments, version control workflows, and ongoing system maintenance
- Participated in code reviews and collaborated with cross-functional teams to deliver features aligned with requirements
- Supported troubleshooting and resolution of production issues in live environments

SOFTWARE DEVELOPER — CODE IT (JUNE 2020 – JULY 2022)

Worked on full-stack development of attendance and reservation systems, focusing on backend logic, databases, and system reliability.

- Developed full-stack features using PHP, JavaScript, and MySQL for enterprise applications
- Built and maintained REST APIs for transactional systems and data-driven workflows
- Designed and implemented relational database structures for scheduling and attendance systems
- Improved performance and usability of legacy systems through refactoring and optimization
- Debugged and resolved production issues to improve system stability and reliability
- Translated requirements into technical solutions in collaboration with stakeholders
- Contributed to incremental improvements and maintenance of existing systems

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COOP EXPERIENCE

LEAP BUSINESS ACCELERATOR — FANSHAWE COLLEGE (MAY 2026 – PRESENT)

Worked in an applied business accelerator environment focused on startup development, product validation, and real-world problem-solving.

- Contributed to startup and product development initiatives in a fast-paced, team-based environment
- Applied structured problem-solving approaches to business and product challenges
- Participated in ideation, validation, and iteration of early-stage concepts
- Collaborated with multidisciplinary teams to execute project-based deliverables
- Developed communication, teamwork, and applied execution skills in entrepreneurial contexts

PERSONAL PROJECT

BEARING FAILURE DETECTION SYSTEM

Developed a machine learning pipeline for predictive maintenance using vibration signal data to detect mechanical bearing failures.

- Built CNN-based deep learning models using TensorFlow and PyTorch for fault classification
- Performed signal preprocessing and feature extraction on vibration datasets
- Designed training pipelines with reproducible experiments and model evaluation workflows
- Applied hyperparameter tuning to improve model performance and stability
- Evaluated model results using standard classification metrics for reliability assessment
- Domain focus: predictive maintenance and industrial fault detection in real-world engineering environments

FATIGUE CLASSIFICATION SYSTEM

Built a computer vision-based classification system for fatigue detection using deep learning models.

- Implemented and compared ResNet, VGG16, and MobileNet architectures using PyTorch
- Developed image preprocessing and augmentation pipelines to improve generalization
- Conducted hyperparameter tuning and model optimization experiments
- Achieved 84% test accuracy on the final trained model
- Built structured evaluation workflow for consistent performance benchmarking
- Designed with consideration for real-world deployment scenarios in safety-critical monitoring systems

GEOSPATIAL ROUTING SYSTEM

Developed a backend system for route optimization and location-based services using Node.js and SQL.

- Built RESTful backend services for geospatial routing and query handling
- Integrated OSRM routing engine for real-time path computation
- Designed database schemas optimized for location-based queries
- Improved query performance through SQL optimization techniques
- Developed backend logic for handling routing requests and responses efficiently
- Designed for scalable deployment in production backend environments

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EDUCATION

GRADUATE CERTIFICATE — ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (CO-OP)

FANSHAW COLLEGE, CANADA | MAY 2025 – AUGUST 2026

Advanced graduate-level program focused on machine learning systems, deep learning, and AI application development with co-op integration.

- Specialized in machine learning, deep learning, computer vision, NLP, and AI system design
- Hands-on experience with PyTorch, TensorFlow, CNNs, transformers, and model training workflows
- Applied statistical methods for model evaluation, optimization, and performance improvement
- Developed practical ML pipelines involving data preprocessing, training, validation, and testing
- Participated in applied projects emphasizing reproducibility and real-world AI system design
- Recognized on Dean's Honour Roll for academic performance

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY

TRIBHUVAN UNIVERSITY, NEPAL | JUNE 2016 – MAY 2020

Built strong foundational knowledge in computer science, software engineering, and mathematical systems used in modern computing.

- Studied data structures, algorithms, databases, operating systems, and computer networks
- Developed understanding of software engineering principles and system design fundamentals
- Strengthened mathematical and statistical foundation for computing and problem solving
- Completed coursework involving programming, systems design, and database management

LEADERSHIP & ACTIVITIES

PICNIC SOCIETY — CLUB PRESIDENT

Led planning and execution of student events and coordinated club activities. Managed member engagement, event organization, and team coordination.

FANSHAW DIGITAL AMBASSADOR

Supported peer students, including international students, by providing guidance on academic systems, digital tools, and campus resources. Assisted in improving student engagement and onboarding experience.

RED SQUAD VOLUNTEER

Participated in event support activities, contributing to coordination, setup, and execution of college events. Demonstrated reliability and teamwork in fast-paced environments.

ADDITIONAL EXPERIENCE

CREW MEMBER — CORA BREAKFAST & LUNCH / FIVE GUYS (SEPTEMBER 2025 – MARCH 2026)

Developed strong customer service, communication, and teamwork skills in fast-paced, high-pressure environments. Managed time-sensitive tasks while maintaining service quality and efficiency.

TELEPHONE INTERVIEWER — LOGIT GROUP (MARCH 2026 – PRESENT)

Conducted structured interviews and collected accurate data for research and survey projects. Strengthened communication, active listening, and attention to detail in professional data collection environments.